

FINANCIAL MARKETS IN DISCRETE TIME

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1. GENERAL IDEAS IN DISCRETE TIME

- **Market model:** Is a tuple $(\Omega, \mathcal{F}, P, \mathbb{T}, \mathbb{F}, S)$ where
 - $(\Omega, \mathcal{F}, P)$ is a probability space.
 - $\mathbb{T} = [0, \dots, T]$ is a time horizon.
 - $\mathbb{F} = \{\mathcal{F}_t\}_{t \in \mathbb{T}}$ is a filtration on $(\Omega, \mathcal{F}, P)$.
 - $\{S_t\}_{t \in \mathbb{T}}$ is an adapted random process $S_t = (S_t^0, S_t^1, \dots, S_t^d)$ with S_t^0 riskless (bond) and S_t^i for $i \geq 1$ are risky assets (stocks). The riskless bond evolves following the deterministic process

$$S_{t+1}^0 = (1 + r)S_t^0.$$

- **Strategy:** A strategy $\{\theta_t\}_{1 \leq t \leq T}$ is a predictable process (with respect to the sequence of filtrations $\mathcal{F}_t$). The strategy represents the sequence of positions of an investor. At time $t - 1$ we take a position and hold this position during the interval $[t - 1, t]$.
- **Portfolio:** The portfolio at time $t \geq 1$ consists of
 - θ_t^0 bonds
 - θ_t^i units of the risky asset S_t^i .

Its value is given by

$$V_t(\theta) = \theta_t \cdot S_t, \quad t \geq 1.$$

The initial investment needed to construct the portfolio is $V_0(\theta) = \theta_1 \cdot S_0$.

- **Put call parity:** derive from simple argument (use linearity of $F = C - P$).

2. ARBITRAGE AND FIRST FUNDAMENTAL THEOREM

It is a **property of the market** itself. We say that there is an arbitrage opportunity in a market if there exists an admissible ($V_t(\theta) \geq 0$) strategy such that

$$V_0(\theta) = 0 \quad \text{and} \quad \mathbb{E}(V_t(\theta)) > 0 \quad a.s.$$

Remark 1. Note that given two equivalent probability measures arbitrage for one if and only if arbitrage for the other.

We say that a market is viable if no-arbitrage opportunities. A probability measure Q is an **equivalent martingale measure** (EMM) if

- $Q \sim P$.
- The discounted process $\bar{S}_t = \beta_t(S_t^1, \dots, S_t^d)$ is a martingale under Q (here $\beta_t = (1+r)^{-t}$).

Remark 2. Example of a non-viable market: binomial with $r < a < b$ (see later).

2.1. The value process of an admissible strategy. We define the **value process** of a strategy θ by

$$V_t(\theta) = \theta_t \cdot S_t$$

Denote by $\bar{S}_t$ the discounted process and let $\bar{V}_t(\theta) = \theta_t \cdot \bar{S}_t$ be the **discounted value process**. If θ is **self-financing**

$$\bar{V}_t(\theta) = \bar{V}_{t-1}(\theta) + \theta_t \Delta \bar{S}_t = V_0(\theta) + (\theta \cdot \bar{S})_t$$

Thus, under a EMM the discounted value process $\bar{V}_t$ of any admissible strategy is a **martingale**.

2.2. The first fundamental theorem of asset pricing. We now relate viability of the market model to the existence of an EMM.

Theorem 2.1 (First fundamental theorem of asset pricing). *In a **finite** market model there exists an EMM if and only if the market is viable.*

We only do the proof for finite market models (Ω is finite dimensional). Can be adapted to general discrete market models. Do not know about continuous markets?

Proof. First notice that viability can be characterized in terms of $\hat{\theta} = (\theta^1, \dots, \theta^d)$ as this portfolio can always be extended to a self-financing $\theta = (\theta^0, \hat{\theta})$ with $V_0(\theta) = 0$. Now viability of the market iff the linear subspace

$$L = \{(\hat{\theta} \cdot \bar{S})_T : \hat{\theta} \text{ is predictable}\} \subset L^2(\Omega),$$

satisfies $L \cap C = \emptyset$ (where C is the (strictly) positive cone). An EMM Q can now be represented by a vector $q \in L^\perp$ with $q_i > 0$ and $|q|_1 = 1$. Indeed we may think of $q \in L^\perp(\Omega)$ as the Radon-Nykodym derivative Q . The fact that $q \in L^\perp$ implies that $\bar{S}$ is a martingale with respect to Q . That this vector exists if $L \cap C = \emptyset$ is an easy consequence of the separation theorem for the vector subspace L and the compact convex set $K = \{\mathbb{E}_P(X) = 1\}$. □

Viability of the market is very relevant when pricing contingent claims whose underlying asset belongs to the market. It is automatic in

- Binomial model
- Continuous time (Girsanov)

What about other models?

2.2.1. Viability of binomial model. If $S_1(\omega_1) = (1+b)S_0$ and $S_1(\omega_2) = (1+a)S_0$ we obtain that

$$\Delta \bar{S}_1(\omega_1) = \frac{b-r}{1+r} \quad \Delta \bar{S}_1(\omega_2) = \frac{a-r}{1+r}$$

We need $\Delta \bar{S}_1(\omega_1) \Delta \bar{S}_1(\omega_2) < 0$ which is equivalent to $a < r < b$.

3. COMPLETENESS AND SECOND FUNDAMENTAL THEOREM

It is again a **property of the market** itself. Let $(\Omega, \mathcal{F}, P, \mathbb{T}, \mathbb{F}, S)$ be a viable market model with a EMM Q . We say that the market is **complete** if, under martingale transform, the discounted process $\bar{S}$ serves as basis for the vector space of $(\mathbb{F}, Q)$ martingales. That is, given any martingale $M \in (\mathbb{F}, Q)$ there exists an adapted process θ such that

$$M_t = M_0 + (\theta \cdot \bar{S})_t.$$

Theorem 3.1 (Second fundamental theorem of asset pricing). *A **finite** viable market is complete if and only if there exists a unique martingale measure.*

Proof. If market complete, for any $\mathcal{F}$ -measurable X there exists $\theta \in \Theta_a$ such that $V_T(\theta) = \beta_T^{-1} X$. But then $\mathbb{E}_Q(X) = V_0(\theta)$ for any EMM Q and the conclusion follows since we can take $X = \mathbb{1}_A$ for all $A \in \mathcal{F}$ and we obtain that $\mathbb{E}_Q(\mathbb{1}_A) = \mathbb{E}_{Q'}(\mathbb{1}_A)$ for any two EMMs Q, Q' .

If market is viable but not complete we show how to construct a family of EMMs. Indeed consider

$$\mathcal{L} = \{\mathbf{c} + (\hat{\theta} \cdot \bar{S})_T : \hat{\theta} \text{ is predictable and } \mathbf{c} = (c_1, \dots, c_n)\}$$

We see that $\mathcal{L}$ is a proper closed subspace of $\mathbb{R}^\Omega$. There exists $Z \in \mathbb{R}^\Omega$ such that $\mathbb{E}_Q(ZY) = 0$ for all $Y \in \mathcal{L}$. Take for example the measure

$$Q'(\omega) = Q(\omega) \left(1 + \frac{Z(\omega)}{2|Z|_\infty}\right)$$

Easy to see that Q' is a different EMM. Convex combinations also form EMM. $\square$

Completeness of the market is of extreme relevance to option pricing and the **construction of hedging strategies** $\longrightarrow$ Law of one price.

Remark 3. Examples of non-complete markets (“trinomial”).

4. ASSET PRICING: GENERALITIES

The first remark is that it is in general **not possible** to assign a unique price to a given contingent claim. This possibility will depend on the viability and completeness of the market.

4.1. **Buyer and seller price.** Given a contingent claim $H \in \mathcal{F}_T$,

- The seller defines its price as the infimum initial investment which allows to construct a hedge

$$\pi_s(H) = \inf\{z : \exists \theta \in \Theta_a \text{ such that } z + G_T(\theta) \geq H\}.$$

- The buyer defines its price as the maximum such that he can construct an strategy such that together with the value of the claim allows him to pay the money borrowed to buy

$$\pi_b(H) = \sup\{z : \exists \theta \in \Theta_a \text{ such that } H + G_T(\theta) \geq z\}.$$

4.2. **General ideas.** In order to “price” a contingent claim depending on an asset:

- **Choose a model:** for the dynamics of the underlying asset. Examples: binomial model, trinomial model, finite differences, continuous time...
- **Calibrate parameters:** For example, in the binomial model set r (interest rate) to match historic data and a, b (up-down factors) to match the volatility (see CRR model).

- **Existence of EMM:** In the binomial model is automatic. In some continuous time models (Black-Scholes) existence is automatic from Girsanov. Finite difference models coming from these continuous time models already implicit as well.

What about other models? Are there popular models (discrete or continuous) in which one has to care about existence and uniqueness of the EMM?

- **Computation of EMM:** In the binomial model or Black-Sholes automatic. In models with non-uniqueness such as trinomial models? Choose based on minimization of the risk function, historical data, intuition, etc...
- **Pricing:** Below we show how to price
 - European claims in discrete complete viable models.
 - European claims in continuous time.
 - American claims in discrete complete viable models.

5. PRICING OF EUROPEAN CONTINGENT CLAIMS

The properties of a market studied above are of relevance to price contingent claims.

5.1. Pricing of European options. The following is the main result in this section.

Theorem 5.1. *Let $H \in \mathcal{F}_T$ be a European contingent claim and let $\mathcal{Q}$ be the set of EMM of the market model. Then,*

$$\pi_s(H) = \sup_{Q \in \mathcal{Q}} \mathbb{E}_Q(\beta_T H) \quad \pi_b(H) = \inf_{Q \in \mathcal{Q}} \mathbb{E}_Q(\beta_T H).$$

Proof. **WRITE** □

5.2. Complete market models: risk neutral pricing. We are given a European contingent claim H . There exists a **unique price** $\pi(H)$ which precludes arbitrage in the extended market model (or equivalent buyer and seller price agree). This price is given by $\mathbb{E}_Q(\beta_T H)$ where Q is the unique EMM for this market.

5.3. Incomplete market models: the arbitrage interval. There is an arbitrage interval

$$\mathcal{I}(H) = [\pi_b(H), \pi_s(H)].$$

5.4. The Binomial model. It is a complete viable market. The **Cox-Ross-Rubinstein** (CRR) formula for option pricing of a European option reads:

$$C_0 := V_0 = S_0 \Psi(A; T, q') - K(1+r)^{-T} \Psi(A; T, q)$$

where $\Psi(m; n, p) = \sum_{j=m}^n \binom{n}{j} p^j (1-p)^{n-j}$. We can also obtain formulas for the hedging strategy.

5.5. From CRR to Black-Scholes model. Consider a binomial model with

- N steps
- Interest rate ρ_N
- Up and down factors a_N and b_N .

This determines uniquely a EMM Q_N . Under this measure the return rates

$$R_n^N = \frac{S_n}{S_{n-1}} \quad \text{are } i.i.d$$

From the CRR formula we obtain that the price of a European put option is given by

$$P_N = \mathbb{E}_{Q_N} \left(K(1 + \rho_N)^{-N} - S_0 \prod R_n^N \right)^+$$

Writing

$$Z^N = \sum_n Y_n^N \quad Y_n^N = \log \left(\frac{R_n^N}{(1 + \rho_N)} \right)$$

so

$$P_N = \mathbb{E}_{Q_N} \left(K(1 + \rho_N)^{-N} - S_0 e^{Z_N} \right)^+$$

Adjusting properly the parameters ρ_N, a_N and b_N one may check that, using the Lindenberg-Feller CLT for triangular arrays Z_N **converges in law** to a normal random variable $\mathcal{N}(-\frac{1}{2}\sigma^2 T, \sigma^2 T)$. This yields the Black-Scholes formula. It is interesting that in this derivation we do not obtain/need any information on the dynamics of the limiting process.

6. PRICING OF AMERICAN CONTINGENT CLAIMS

Definition 6.1. An American option with payoff process $\{f(S_t)\}_{t \leq T}$ and expiration T gives the holder the possibility to **exercise** it **at any** $t \in \{0, \dots, T\}$ and receive a payoff $f(S_t)$.

Suppose Q is the EMM of the market model and let $f(S_t)$ be the payoff of the claim at time t .

The idea behind option pricing is more involved than for European claims. For American options:

- Construct the Snell envelope Z_t of the payoff process. It dominates the payoff process.
- Z_t is a supermartingale $Z_t = M_t - A_t$. It is a martingale up to the largest optimal stopping time¹.
- Construct a portfolio θ which replicates M_T ². Clearly $M_t (= \bar{V}_t(\theta))$ dominates the payoff process.
- Exercise is only optimal at optimal stopping times because at not optimal times $\bar{f}(S_t) < Z_t \leq M_t$ (if we exercise we receive $f(S_t)$ so we could instead have not bought the option at $t = 0$ but sold it and construct a portfolio with discounted value $\bar{V}_t(\theta)$).

For European options:

- Just construct the portfolio which replicates the random variable $f(S_T)$.

Definition 6.2. Given a process X_t we define its **Snell envelope** Z_t to be the process

$$Z_t = \max\{X_t, \mathbb{E}_Q(Z_{t+1} | \mathcal{F}_t)\} \quad Z_T = X_T.$$

Remark 4. Notice that in general, the fact that S_t is a martingale does not imply that $f(S_t)$ is a martingale (see below Sections 6.2 and 6.3).

The following is the main result of this section.

Theorem 6.3. Let $\{f_t(S_t)\}$ denote the payoff process of an American option and define Z_t to be the Snell envelope of the discounted payoff process $P_t = \beta_t f(S_t)$. Let $\mathcal{Q}$ be the set of EMM of the market model and let $\mathcal{T}$ be the set of stopping times. Then, **THINK AND PROVE**

$$\pi_s(H) = \sup_{Q \in \mathcal{Q}} \sup_{\tau \in \mathcal{T}} \mathbb{E}_Q(Z_\tau) \quad \pi_b(H) = \inf_{Q \in \mathcal{Q}} \inf_{\tau \in \mathcal{T}} \mathbb{E}_Q(Z_\tau).$$

¹Notice that A_t will always be zero if $f(S_t)$ is a submartingale (for example for American calls). This explains why in American call option there is no benefit from early exercise.

²Notice that M_T is in general different from $Z_T = \bar{f}(S_T)$ though.

6.1. Proof of Theorem 6.3. PROVIDE DIFFERENT PROOF

Lemma 6.4. *The Snell envelope is the smallest supermartingale which dominates X_t .*

Definition 6.5. A stopping time σ is **optimal** if it satisfies

$$\mathbb{E}(X_\sigma) = \sup_{\tau} \mathbb{E}(X_\tau).$$

Lemma 6.6. *Let Z_t be the Snell envelope of X_t . The stopping time $\tau_* = \min\{t \geq 0: X_t = Z_t\}$ is optimal for X_t (think of X_t as the discounted payoff) and*

$$Z_0 = \mathbb{E}(X_{\tau_*}).$$

The following is key to the analysis of option pricing for American claims.

Proposition 6.7. *The stopped process $Z_{t \wedge \tau_*}$ is a martingale (recall Z_t is a submartingale only).*

Proof. Notice that while $\tau_* \geq t$ we have that $Z_t = \mathbb{E}(Z_{t+1}|\mathcal{F}_t)$. □

Remark 5. Discuss also the largest optimal stopping time.

6.1.1. *Construction of a minimal hedge.* Let Z_t be the Snell envelope of the discounted payoff $\bar{f}_t(S_t) = (1+r)^{-t}f_t(S_t)$. Since Z_t is a supermartingale it admits a **Doob decomposition** as

$$Z_t = M_t - A_t$$

with M_t a martingale and A_t a predictable increasing process. Since market is viable, there exists $\theta \in \Theta_a$ such that $V_T(\theta) = M_T(\theta)$. We need

$$x = V_0(\theta) = \sup_{\tau} \{\mathbb{E}((1+r)^{-\tau}f_{\tau}(S_{\tau}))\}$$

as initial investment to construct this hedge.

6.1.2. *Optimal exercise.* Exercise is only optimal at optimal times ($\tau_* \leq t \leq \nu$, with ν the largest optimal time).

6.2. American call options.

Theorem 6.8 (No benefit from early exercise). $\tau_* = T$ is an optimal exercise time for an American call option.

Proof. Denote by $P_t = \beta_t f(S_t)$. Observe that since $\bar{S}_t$ is a martingale and the function $S \mapsto (S - K)^+$ is convex

$$\mathbb{E}_Q(P_{t+1}|\mathcal{F}_t) = \mathbb{E}_Q((\bar{S}_{t+1} - \beta_{t+1}K)^+|\mathcal{F}_t) \geq (\bar{S}_t - \beta_{t+1}K)^+ \geq \beta_t(S_t - K)^+ = P_t.$$

We conclude that the Snell envelope is just $Z_t = \mathbb{E}_Q(P_{t+1}|\mathcal{F}_t)$ for all $t = T, \dots, 1, 0$. □

6.3. **American put options.** In this case, if we try to reproduce the argument in Theorem 6.8 we obtain as well

$$\mathbb{E}_Q(P_{t+1}|\mathcal{F}_t) = \mathbb{E}_Q((\beta_{t+1}K - \bar{S}_{t+1})^+|\mathcal{F}_t) \geq (\beta_{t+1}K - \bar{S}_t)^+$$

but the last term is smaller or equal than $P_t = \beta_t(K - S_t)^+$. In particular, for in the money options we might have that

$$P_t = \beta_t(K - S_t)^+ > \mathbb{E}_Q(P_{t+1}|\mathcal{F}_t).$$

7. PRICING OF CONTINGENT CLAIMS IN NON COMPLETE AND/OR VIABLE MODELS

Talk about risk function. Minimize risk.

8. THE TRINOMIAL MODEL

Idea: take trinomial model. Find an equivalent EMM (these are like the one below with $p \in (0, 1/2)$). Fix that $p \in (0, 1/2)$. Then

- For the corresponding process Donsker's thm shows convergence to BM with p -dependent probability.
- Also, we can find a formula for the price using the CLT (as in binomial model). Price formulas match.
- We must see that in the limit there is a replicating portfolio.

We divide $[0, 1]$ into N intervals of size $\Delta t = 1/N$. We consider the discrete process $\{S_k\}_{k=0}^N$ in which

$$\Delta S_k = \sigma \sqrt{\Delta t} \xi_k$$

where ξ_k are i.i.d. r.v. taking the values $\{-1, 0, 1\}$ with

$$\mathbb{P}\{\xi_k = 1\} = p = \mathbb{P}\{\xi_k = -1\}.$$

Define $\mathcal{F}_i = \sigma(S_k, k \leq i)$ and let $h \in \mathcal{F}_N$ be a contingent claim. We want to study the value process

$$V_k = \mathbb{E}(h | \mathcal{F}_k).$$

Clearly V_k is a martingale. However, we know that since the market is not complete, it is a priori not possible to express V_k as a martingale transform. Here we show that the lack of completeness vanishes in the limit $\Delta t \rightarrow 0$ provided h is a C^2 function. To do so we observe that

$$e_1 = \frac{\xi_k}{\sqrt{\text{var} \xi_k}} = \frac{1}{\sqrt{2p}} \xi_k \quad e_2 = \frac{\xi_k^2 - \mathbb{E} \xi_k^2}{\sqrt{\text{var}(\xi_k^2 - \mathbb{E} \xi_k^2)}} = \frac{1}{\sqrt{2p(1-2p)}} (\xi_k^2 - 2p)$$

form an orthonormal basis of $L_0^2 \simeq \mathbb{R}^2$, the space of functions which are measurable with respect to $\sigma(\xi_k)$, have zero mean and are square summable. Then, if we define

$$\Delta V_k = \mathbb{E}(h | \mathcal{F}_k) - \mathbb{E}(h | \mathcal{F}_{k-1})$$

a straightforward computation shows that at any $\omega \in \{-1, 0, 1\}^k$

$$\begin{aligned} \langle \Delta V_k, e_1 \rangle(\omega) &= \sqrt{\frac{p}{2}} (V_k(\sigma \sqrt{\Delta t}, \omega) - V_k(-\sigma \sqrt{\Delta t}, \omega)) \\ \langle \Delta V_k, e_2 \rangle(\omega) &= \sqrt{\frac{p(1-2p)}{2}} (V_k(\sigma \sqrt{\Delta t}, \omega) - 2V_k(0, \omega) + V_k(-\sigma \sqrt{\Delta t}, \omega)). \end{aligned}$$

Hence, we may write

$$\begin{aligned} \Delta V_k &= \langle \Delta V_k, e_1 \rangle e_1 + \langle \Delta V_k, e_2 \rangle e_2 \\ &= \frac{1}{2} (V_k(\sigma \sqrt{\Delta t}) - V_k(-\sigma \sqrt{\Delta t})) \xi_k + \frac{1}{2} (V_k(\sigma \sqrt{\Delta t}) - 2V_k(0) + V_k(-\sigma \sqrt{\Delta t})) (\xi_k^2 - 2p) \\ &= H_k \Delta S_k + G_k (\Delta S_k^2 - 2p \sigma^2 \Delta t). \end{aligned}$$

where

$$H_k = \frac{V_k(\sigma \sqrt{\Delta t}) - V_k(-\sigma \sqrt{\Delta t})}{2\sigma \sqrt{\Delta t}} \quad G_k = \frac{1}{2} \frac{V_k(\sigma \sqrt{\Delta t}) - 2V_k(0) + V_k(-\sigma \sqrt{\Delta t})}{\sigma^2 \Delta t}.$$

Summing over k , for each $0 < n \leq N$ we have

$$V_n = V_0 + \sum_{k=0}^n \Delta V_k = \mathbb{E}h + \sum_{k=0}^n H_k \Delta S_k + \sum_{k=0}^n G_k (\Delta S_k^2 - 2p \sigma^2 \Delta t).$$

APPENDIX A. STOPPING TIMES

Let τ be a stopping time. We define the information available at time τ (see also figure ... for a set not in $\mathcal{F}_\tau$)

$$\mathcal{F}_\tau = \{A \in \mathcal{F} : A \cap \{\tau \leq t\} \in \mathcal{F}_t\}.$$